Midterm 2 Review Guide
STAT 132; Winter 2025

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1 Sampling Distributions

- Sampling Distribution: Let $T = t(X_1, X_2, \ldots, X_n)$ be a statistic. Then, the probability distribution of T is called the **sampling distribution** of T.
  - The sampling distribution provides insight into the variability of a statistic.
  - The standard deviation of the sampling distribution is called the **standard error**.
  - As the sample size n increases, the sampling distribution of many statistics become closer to a normal distribution.
  - As the sample size increases, the sampling distribution of many statistics have lower variance.
- Be able to find the sampling distribution for different MLEs such as the mean of a normal distribution, the upper bound of a uniform distribution, etc.

2 The χ^2 Distribution

- A continuous random variable X is said to follow a chi-square distribution with m degrees of freedom if it has the following pdf:

$$f(x|m) = \frac{1}{2^{m/2}\Gamma(m/2)} x^{(m/2)-1} e^{-x/2}, \quad x > 0.$$

  This is denoted as $X \sim \chi^2_m$.
- **Mean:** $E(X) = m$.
- **Variance:** $\text{Var}(X) = 2m$.
- **Additivity:** If $X_1 \sim \chi^2_{m_1}$ and $X_2 \sim \chi^2_{m_2}$, and X_1 is independent of X_2, then $X_1 + X_2 \sim \chi^2_{m_1+m_2}$. Conversely, if $X_3 \sim \chi^2_{m_1+m_2}$ with $X_1 \sim \chi^2_{m_1}$, $X_2 \sim \chi^2_{m_2}$, and $X_3 = X_1 + X_2$, then X_1 and X_2 must be independent.
- If Z has a standard normal distribution, then Z^2 has a χ^2_1 distribution.
- More generally, the chi-square distribution is the sum of independent squared standard normal variables: $Z_1^2 + Z_2^2 + \cdots + Z_m^2 \sim \chi^2_m$.
- If $X \sim \chi^2_m$, then, for some constant c, the random variable $Y = cX$ will have a Gamma$(\frac{m}{2}, \frac{1}{2c})$ distribution.
- Know how to use the χ^2 table to determine probabilities or quantiles.

3 Sample Mean and Sample Variance for a Normal Population

Suppose $X_1, X_2, \ldots, X_n$ are i.i.d. random variables from $N(\mu, \sigma^2)$. Then:

- The sample mean, $\hat{\mu} = \bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$, follows a normal distribution:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

- The sample variance, $\widehat{\sigma^2} = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$, satisfies:

$$\frac{n\widehat{\sigma^2}}{\sigma^2} \sim \chi^2_{n-1}.$$

- $\bar{X}$ and $\widehat{\sigma^2}$ are independent.

4 The t Distribution

- Let $Z \sim N(0,1)$ and $Y \sim \chi^2_m$, where Z and Y are independent. The random variable

$$T = \frac{Z}{\sqrt{Y/m}} \tag{1}$$

  follows a t-distribution with m degrees of freedom (denoted as $T \sim t_m$).

- This distribution is symmetric about 0.
- For $m = 1$, the mean does not exist. For $m > 1$, the mean is zero.
- For $m > 2$, $\mathrm{Var}(X) = \frac{m}{m-2}$.
- As $m \to \infty$, the t-distribution converges to the standard normal distribution.
- The PDF of the t distribution with m degrees of freedom is

$$f(x|m) = \frac{\Gamma\left(\frac{m+1}{2}\right)}{\sqrt{m\pi}\,\Gamma(m/2)}\left(1 + \frac{x^2}{m}\right)^{-(m+1)/2}, \quad -\infty < x < \infty.$$

- Suppose that $X_1, \ldots, X_n$ form a random sample from a normal distribution with mean μ and variance σ^2. Let $\bar{X}$ denote the sample mean and define

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

  Then, the quantity

$$U = \frac{\sqrt{n}(\bar{X} - \mu)}{\sqrt{S^2}}$$

  will follow the t distribution with $n - 1$ degrees of freedom.

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- Know how to use the t table to determine probabilities or quantiles.

5 Interval Estimation

- Let $X_1, \ldots, X_n$ be a random sample from some population that depends on the parameter θ. Let L and U be two statistics calculated from the data (with $L < U$). Then if

$$\mathbb{P}(L < \theta < U) = 1 - \alpha$$

the **random interval** (L, U) is called a $1 - \alpha$ confidence interval. For example, if $\alpha = 0.05$, this would result in a 0.95 or 95% confidence interval.

- Let $X_1, X_2, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with known σ. Then a $1 - \alpha$ confidence interval for μ can be constructed as

$$\bar{X} \pm z_{1-\alpha/2} \cdot \frac{\sigma}{\sqrt{n}},$$

where $z_{1-\alpha/2}$ is the $1 - \alpha$ quantile of the standard normal distribution.

- Let $X_1, X_2, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with unknown σ. Then a $1 - \alpha$ confidence interval for μ can be constructed as

$$\bar{X} \pm t_{n-1,1-\alpha/2} \cdot \frac{S}{\sqrt{n}}.$$

where $t_{n-1,1-\alpha/2}$ is the $1 - \alpha$ quantile of the t-distribution with $n - 1$ degrees of freedom, and

$$S = \sqrt{S^2} = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2}.$$

- Know how to find confidence intervals for other parameters such as the variance of a normal distribution.

- If $Q(\boldsymbol{X}, \theta)$ is a function of the data/parameter with a distribution does not depend on θ (i.e., the distribution is the same for any value of θ), then $Q(\boldsymbol{X}, \theta)$ is called a pivotal quantity.

- Understand the long-run frequency interpretation of the confidence interval.